

Analysis

Revamping Europe's chips strategy: indispensability, not self-sufficiency

In revising the European Chips Act, the EU should seek to make itself an indispensable partner in the semiconductor value chain

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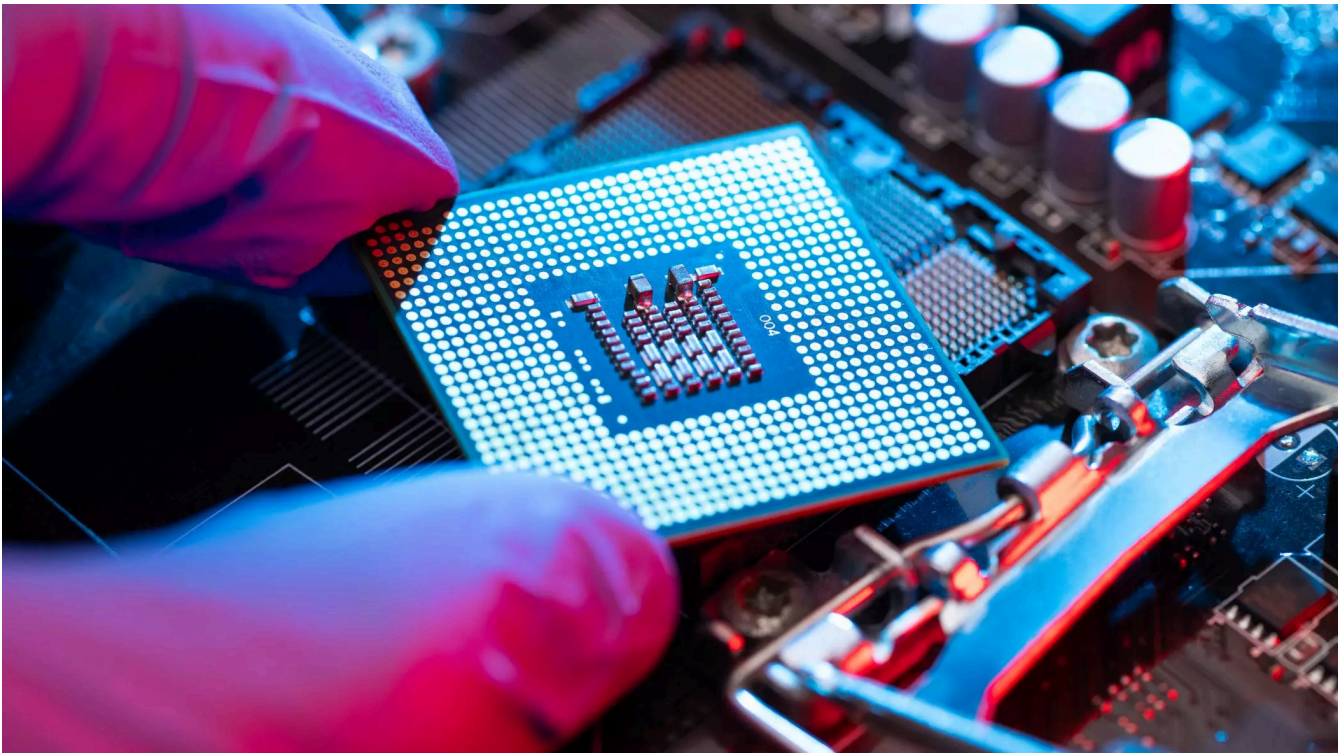
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Adopted in 2023 in the wake of pandemic-era supply-chain shocks, the European Chips Act (Regulation (EU) 2023/1781) was intended to facilitate an expansion of the European Union's semiconductor industry. However, in terms of the scale of

The European Commission now plans a Chips Act 2. The proposed revision¹ must address the Chips Act's shortcomings along with the new demands of the artificial-intelligence era and the shifting geopolitical reality that have emerged since it took effect. It remains necessary to reduce critical vulnerabilities but policy should not seek to replicate entire chip ecosystems in pursuit of self-sufficiency. The Chips Act 2 should pursue technological sovereignty through indispensability by building on and defending European strengths in semiconductors.

Taking stock of the Chips Act

The European Chips Act has a headline target of raising Europe's share of global production of advanced semiconductors to 20 percent by 2030². It has pursued this goal by:

1. Providing EU funding for semiconductor research, including initiatives that bridge the gap between laboratory advances and industrial-scale production (pillar I);
2. Establishing a framework to support semiconductor manufacturing capacities if they are novel in Europe (the so-called 'first-of-a-kind' rule; European Commission, 2022) (pillar II);
3. Establishing the European Semiconductor Board, which brings together EU country representatives and the European Commission to monitor the semiconductor value chain and coordinate crisis response (pillar III).

The Chips Act has been criticised for its lack of feasibility and focus (Poitiers and Weil, 2022). This can be partly blamed on the Commission's limited mandate. The Act aims to stimulate €43 billion of public investment, but the Commission is only responsible for €4.5 billion of that (disbursed under pillar I). Support for novel semiconductor manufacturing (pillar II) is a job for state aid decided on by EU governments, with the Commission having only very limited authority in coordinating such aid (ECA, 2025).

Furthermore, global competition to attract semiconductor manufacturing is fierce and costly, aggravated by an ongoing subsidy race (OECD, 2025). The United States's CHIPS and Science Act provided the US Department of Commerce with large funding and substantial discretion to act³. As of January 2025, it had awarded \$33.7 billion in grants and \$5.5 billion in loans for state aid (Commerce

approved⁴.

The Chips Act also relies heavily on ambitious supply-side measures, which do not always correspond to the structure of European demand (Poitiers and Weil, 2022a; Kleinhans, 2024). Semiconductor demand in Europe comes primarily from the automotive sector and industrial applications, which do not rely on cutting-edge chips (ECA, 2025).

The relative lack of success of the Chips Act measured against its objectives is illustrated by Intel's July 2025 cancellation of a plan to build an advanced semiconductor fabrication plant in Magdeburg, Germany. Though initially conceived as the flagship pillar II investment, Intel CEO Lip-Bu Tan cited insufficient customer commitments as a main factor behind the cancellation⁵.

Emerging fault lines

The Chips Act 2 must also address challenges that have emerged since the first Chips Act was proposed in February 2022. Demand for AI chips has surged and the trajectory shows no sign of flattening. Hyperscale investment in data centres is reshaping the demand for advanced semiconductors, which are used in AI training and inference. The shift is unlikely to be cyclical. It is underpinned by so-called 'scaling laws': that more computing power leads to better AI model performance (Sevilla *et al*, 2022).

The rapidly growing strategic and economic significance of AI has turned the chip supply chain into a geopolitical asset. The US national security strategy now treats semiconductors as core security infrastructure⁶. In October 2022, a month before OpenAI released ChatGPT, Washington introduced export controls with the explicit objective of constraining Chinese AI development, framed as a matter of national security⁷. Europe is not a bystander in this contest: it has influence over several of the potential chokepoints the US strategy depends on.

The most critical of these is ASML. This Dutch firm has a monopoly on extreme ultraviolet (EUV) lithography equipment, required for leading-edge semiconductor production. Washington has repeatedly lobbied the Dutch government to revoke export licences for ASML shipments to China⁸. A US bill under consideration, the Multilateral Alignment of Technology Controls on Hardware (MATCH) Act, would prohibit ASML from selling its lithography systems to China altogether. This could materially damage ASML's commercial position as the Chinese market accounted for 33 percent of its sales in 2025⁹.

its security strategy, which prefers bilateral engagement with EU countries in the interest of US policy goals (García de Viedma, 2026).

Developments in autumn 2025 also illustrated the rising geopolitical pressure on the semiconductor industry. In summer 2025, Nexperia, the Dutch-based, Chinese-owned producer of mature-node chips, began to transfer production know-how from the Netherlands to China and channelled outsized orders to prop up the struggling Chinese firm WingSkySemi¹⁰. In response, the Dutch government froze the company's assets and a Dutch court subsequently removed its Chinese-appointed CEO. Beijing retaliated by blocking exports to the EU of Nexperia chips manufactured as wafers in Hamburg and shipped to Dongguan for cutting and packaging¹¹. The resulting panic in the European automotive industry led the Dutch government to suspend its intervention¹².

From self-sufficiency to indispensability

It remains to be seen how the Chips Act 2 might confront these challenges. The emerging strategy points towards sovereignty through self-sufficiency. European Commissioner for Technological Sovereignty, Security and Democracy Henna Virkkunen has said the revised act will double down on the ambitions of the first, seeking to onshore design and manufacturing of both AI and mature chips in Europe¹³. France, Poland, Austria and Lithuania have gone further, suggesting that European demand for advanced semiconductors could soon be met, at least partly, by European production¹⁴.

Pursuing self-sufficiency across the chip value chain risks repeating the Chip Act's central mistake: dispersing scarce public resources in a bid to achieve unattainable targets. Leading-edge chip production in the EU is unrealistic in the near future. High capital costs are only part of the reason. Chip production also requires a dense supply ecosystem, which has developed over decades in Asia. Moreover, chip-producing nations Taiwan and Korea depend on tacit process knowledge that resides in a small global pool of experienced engineers; those countries have little strategic interest in sharing technology.

Moreover, the focus on fabs, or production plants, mistakes a part of the value chain for the whole. The Nexperia example of front-end wafer production in Europe but back end cutting, packaging and testing done in China shows Beijing has substantial leverage despite the European location of the fabs. Even if Europe could replicate large chunks of the value-chain domestically, it is far from clear that European chip consumers would pay the substantial premium for these chips.

'sovereignty through indispensability' should be pursued: controlling inputs, capabilities or chokepoints that other players cannot route around.

This approach has attracted support from EU countries¹⁵ and industry¹⁶. Indispensability translates into sovereignty through multiple mechanisms. First, it gives Europe some influence when other powers want to take action. US export controls on lithography tools to China, for instance, only function with Dutch cooperation. Second, it provides Europe with a credible retaliation threat, should others weaponise their own chokepoints. Third, it creates reciprocal dependence: when others need European inputs, they acquire their own commercial interest in keeping Europe stable, supplied and integrated into the global system.

Pursuing sovereignty through indispensability requires a move beyond simplistic targets. State aid remains valuable, but only when deployed in a disciplined, coordinated and strategic manner. The European semiconductor ecosystem should be mapped in detail to identify where Europe controls chokepoints that must be defended, where European strengths can be developed into new chokepoints and where critical dependencies need reducing.

The Chips Act tasked the European Semiconductor Board with monitoring the supply chain for crisis events, but the Board does not coordinate industrial policy in the way envisaged here. Nor would the Commission have the mandate to do so. Dedicated capacity is similarly thin in EU countries. Even in countries with significant semiconductor industries, few officials work exclusively on the sector; most cover several technologies in parallel (Kleinhans, 2024). An indispensability-oriented strategy demands more resources.

Because European chip consumers will need to pay a premium for domestically produced chips, demand-side measures that channel demand towards European producers and trusted partners are increasingly debated as a complement to supply-side support¹⁷. The Chips Act 2 could make public subsidies and access to government procurement conditional on sourcing important semiconductor components from European and trusted-partner suppliers. However, such policies come with substantial administrative burdens, are heavily distortive and might be ineffective in markets in which public procurement plays a limited role, such as the semiconductor market. Therefore, they should be considered with caution.

A sovereignty through indispensability strategy will also require instruments beyond industrial policy and the Chips Act 2. The EU needs to coordinate a coherent response to US and Chinese semiconductor policy. Export controls are

equipment, but operational competence under the law remains national.

EU-level export controls would offer the advantage of distributing countermeasures across the EU and reducing vulnerability to external pressure on any single member state. However, this approach would face the structural obstacles that export controls require intelligence, which is held by national services, and proximity to the companies under review, which Brussels often lacks. A hybrid model comparable to EU legislation on screening foreign direct investment (the FDI Screening Regulation, Regulation (EU) 2019/452) might be workable. This preserves national decision-making while building an EU-wide architecture for information-sharing, mutual consultation and Commission opinions on cross-border cases¹⁸.

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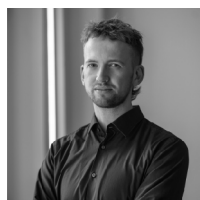
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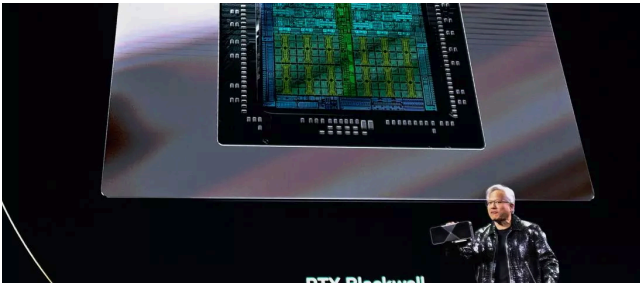
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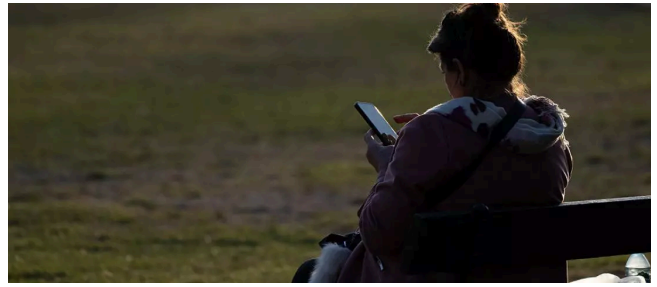
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